Fiscal Year:	FY 2008	**Task Last Updated:**	FY 07/14/2008
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Project Title:	NSCOR: Progressive Alterations of Central Nervous System Structure and Function Are Caused by Charged Particle Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92350-1700	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCOR 03-OBPR-02
Start Date:	10/01/2003	**End Date:**	11/28/2009
No. of Post Docs:	8	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date extended via NCE to 11/28/2009 per PI (3/2009) End date changed to 2/28/2009 per PI (5/08) End date changed to 1/23/2009 per JSC info update (7/07)		
Key Personnel Changes/Previous PI:	Weili Lin, Ph.D. University of North Carolina. Subcontract terminated January 23, 2007. ; Marco Stampanoni, Ph.D. Paul Scherrer Institute, Villigen, Switzerland. New subcontract ; Lead Investigator for X-ray Tomaography. New Subcontract November, 2006.		
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Grant/Contract No.:	NNJ04HC90G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The experimental program detailed above will provide a comprehensive, quantitative assessment of the effects of charged particle radiation on the mammalian brain using the mouse hippocampus as a model. The mouse brain will be irradiated with accelerated protons and iron ions and examined over the course of two years for progressive changes in hippocampal structure and function. Cell populations in tissue sections will be quantified by a high precision stereology method to measure rates of loss of neurons, vascular endothelial cells and glia. Brain blood vessel topology will be further characterized by an erosion casting technique that produces a 3-D map of the vessels at 1 micron resolution. Stem cells and ongoing neurogenesis will be quantified using a complementary technique in which dividing cells are directly detected. Activation of cells and residual genetic damage in these cells will be characterized by immunofluorescence and mRNA level measurements of selected cytokine biomarkers and analysis of mutations in the transgene b-galactosidase. Functional impairments of neuronal ensembles will be quantified by electrophysiological measurements of hippocampal tissue slices and patch clamp measurements will detail changes in membrane properties. System level changes will be quantified by electroencephalography and two magnetic resonance imaging modalities that map white matter tracts (diffusion tensor imaging) and local metabolic rate (oxygen extraction). The plasticity of the brain as a system will be tested in its reaction to an immune system challenge mediated by lipopolysacharide. Finally the ability of radiation to modulate the latency of a neurodegeneration process will be tested in a transgenic mouse that overproduced amyloid precursor protein and develops an Alzheimer disease-like pathology. Taken together these results should provide a basis for determining risk to the CNS from charged particle radiation exposure. The main hypothesis for the grant is: Charged particle radiation exposure induces progressive deleterious structural and functional changes in the brain, leading to performance decline. Research Aims: Hypothesis (A) Charged particle irradiation causes a progressive loss of cells and a remodeling of CNS tissue as a function of dose, time and LET. To test hypothesis A we will: Specific Aim A1. Quantify the cellular composition of mouse brain vascular endothelium using stereological analysis of preserved tissue. Specific Aim A2. Quantify the composition of mouse neurons, selected glia and stem cells using stereological analysis of preserved tissue. Specific Aim A3. Quantify the activation of resident and infiltrating mouse immune cells using stereological analysis immune-labeled tissue. Specific Aim A4. Quantify key molecular biomarkers associated with radiation-induced damage to mouse cells and their microenvironment using targeted gene expression profiling and mutation analysis of a structural gene. Hypothesis (B) Charged particle irradiation alters the functional output or performance of the brain as a function of dose, time and LET. To test hypothesis B we will: Specific Aim B1. Quantify tissue-level electrophysiological properties of mouse brain using tissue slices that represent intact regional ensembles of interacting cells. (extracellular, patch clamp) Specific Aim B2. Map the time evolution of macroscopic structural features and metabolic alteration in mouse brains using MRI. Hypothesis (C) Exposure to radiation decreases disease latency and alters response to acute stressors. To test hypothesis C we will: Specific Aim C1. Quantify the magnitude and time course of electrophysiological changes in brain slices of mice expressing a human transgene that produces an Alzheimer disease-like pathology and limits output. Specific Aim C2. Quantify the electrophysiological performance of brain slices in mice reacting to the application of systemic immune stress using lipopolysaccharide as a surrogate infectious agent Foremost in the Critical Path Roadmap system for radiation effects on the central nervous system is Risk #29 – Acute and Late CNS Risks. All of the research and technology questions associated with this risk have priority level one for Mars missions and priority level 3 levels for ISS and 30d lunar missions. The questions addressed by the NSCOR project are 29a (functional changes), 29b (late degeneration), 29c (Alzheimer predisposition), 29e (late degeneration), 29g (cell and tissue mechanisms of damage), 29h (different cell populations including stem cells) and 29i (biomarkers). Also, a NASA workshop on CNS radiobiology identified supporting questions to assist in designing experiments to elucidate CNS effects in response to radiation exposure. These questions include: 1) What is the latency to CNS injury, 2) What cell types are involved in CNS injury (neurons, stem cells, glia)? and 3) What vascular changes occur in response to exposure. We have used this framework to focus our investigation on the effects of charged particles on the CNS. Specifically we will quantify cellular and functional changes in the hippocampus following high energy protons (main SPE constituent) and Fe-56 particles . We will use male rodents with sufficient statistical samples to address inter-individual variation and will address workshop questions 1-3 by examining the responses of different constituent cell types and the latency of damage. NSCOR subproject A quantifies different cell populations (29b, 29e, 29g, 29h); subproject B addresses functional changes and non-invasive techniques for identifying biomarkers (29a, 29i); and subproject C targets late degenerative changes with an Alzheimer model and the role of inflammation (29b, 29c, 29e, 29g and 29i). Our strategy is therefore aligned with the programmatic goals set forth by NASA and its advisory groups.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The response of the central nervous system to ionizing radiation will have direct impact on the development of radiotherapy using charged particles. The ability of protons, helium ions and carbon ions to be localized precisely in treatment fields close to critical structures (eg. optic nerve) make charged particle radiotherapy highly desireable. The limiting biological constraint on such treatments is the reaction of normal tissue surrounding a tumor site. The results of the NSCOR task will quantify cell loss and tissue functional changes form proton and iron irradiation that can be used directly to estimate normal tissue complication probabilities in the context of radiotherapy.
	This NASA program project investigates the responses of a mammalian brain structure (hippocampus) to charged-particle radiation. Effects of protons and iron ions are being compared for up to two years post irradiation in C57Bl/6 mice and two transgenic variants. The first project aim is to quantify time- and dose-dependent changes in neurons, glia, stem cells and vascular endothelium using stereological analysis of tissue sections. A polymer infusion technique complements this analysis and maps vasculature topology by X-ray tomography. The second aim is to quantify functional status. The primary measurements are extracellular electrical recordings from brain slices, and patch clamp measurements of nerve membrane properties. MRI sequences map fiber tracts for structural integrity. Brain's ability to maintain homeostasis is assessed following a controlled immune shock elicited by lipopolysaccharide injection and a transgenic mouse that develops Alzheimer's disease-like pathology is used to determine whether radiation exposure alters the latency and severity of the neurodegeneration process. The third overall aim is to quantify molecular

markers underlying cellular and system changes. For this the team quantifies the frequency and spectrum of ß-galactosidase reporter gene mutations and employs transcription profiling to assess the status of genes associated with cytokine signaling and synaptic plasticity. In this project, the time evolution of CNS reactions is observed from one month onward and does not characterize the highly dynamic changes occurring immediately after exposure.
Space Radiation

The unique feature of the space radiation environment is the presence of high-energy charged particles dominated by protons from the sun and a complex spectrum of galactic cosmic rays consisting of ionized nuclei of stable elements. Of these, protons are estimated to contribute 19% of the dose equivalent in space while 56Fe ions constitute 13% and carbon and oxygen each contribute about 6 %. Estimated radiation exposures for long term missions are estimated to be of the order of 1 – 2 Sieverts.

CNS Radiobiology

The brain is the site of executive control over homeostatic mechanisms via electrical and endocrine signaling networks, and therefore of critical importance to all other organ and tissue systems under its coordination. While high radiation doses to the brain result in significant tissue destruction (e.g. demyelination and necrosis), subtle changes can occur that may have significant physiologic impact. Such changes could involve loss of critical cell populations such as neural precursor cells, neurons, and glia or alterations in the microvasculature. Microenvironment changes might result in dysregulation of synaptic plasticity and electrical signal processing. Little data are available radiation reactions of CNS at low doses and with charged particles which will determine the risks associated with protracted charged particle exposures in space.

Hypotheses

With these CNS and radiation features in mind, we established a main hypothesis and a set of three specific hypotheses to organize our experimental strategy. The main hypothesis is: Charged particle radiation exposure induces progressive deleterious structural and functional changes in the brain, leading to performance decline. This leads to the following specific hypotheses and associated aims:

Hypothesis (A) Charged particle irradiation causes a progressive loss of cells and a remodeling of CNS tissue as a function of dose, time and LET. To test hypothesis A we: Specific Aim A1. Quantify the cellular composition of mouse brain vascular endothelium using stereological analysis of preserved tissue. Specific Aim A2. Quantify the composition of mouse neural precursor cells using stereological analysis of preserved tissue. Specific Aim A3. Quantify the activation of microglial cells using stereological analysis immune-labeled tissue. Specific Aim A4. Quantify key molecular biomarkers associated with radiation-induced damage using targeted gene expression profiling and mutation analysis of a structural gene.

Hypothesis (B) Charged particle irradiation alters the functional output or performance of the brain as a function of dose, time and LET. To test hypothesis B we: Specific Aim B1. Quantify electrophysiological properties of mouse brain using tissue slices that represent intact regional ensembles of interacting cells. (extracellular, patch clamp) Specific Aim B2. Map the time evolution of macroscopic structural features in mouse brains using Magnetic Resonance Imaging.

Hypothesis (C) Exposure to radiation decreases neurodegenerative disease latency and alters the response to acute stressors. To test hypothesis C we: Specific Aim C1. Quantify the magnitude and time course of changes in brains of mice manipulated by a mutation that produces an Alzheimer disease-like pathology and limits output (electrophysiology, Aß plaque load, vessel topology). Specific Aim C2. Quantify the electrophysiological performance of brain slices in mice reacting to the application of systemic immune stress using lipopolysaccharide as a surrogate infectious agent.

Animal Models

The male C57/BL6 mouse is being used for experiments. It was chosen based on its suitability for specific experimental protocols, general reliability in predicting human responses, and wealth of published radiobiological data. Two genetic variants of the C57/BL6 mouse will be used in addition to wild type: an APP (amyloid precursor protein) transgenic strain, APP 23 tg locus and a transgenic variant containing an integrated shuttle vector array with the E. coli ß-galactosidase gene (lacZ).

Irradiation Considerations

To investigate the contributions of the two most important and qualitatively different particles, NSCOR experiments utilize doses of 0.5 to 4 Gy for 56Fe ions and 0-8 Gy for protons. We selected: 250 MeV/amu 1H1+ (LET 0.4 keV/µm) and 600 MeV/amu 56Fe26+ (LET 175 keV/µm). The 250 MeV energy for protons is typical of space radiation and for radiotherapy at Loma Linda University where proton beams are obtained. For iron, 600 MeV/amu was selected to maximize LET, while minimizing fragmentation and facilitating use of a collimator to provide brain-only irradiation. Iron beams are obtained from the NASA Space Radiation Laboratory (NSRL) at the Brookhaven National Laboratory.

Task Progress:

Key Observations and Significance

Aim A1

Significant time and dose dependent progressive loss of endothelial cells are seen along with loss and remodeling of microvessels; hippocampal subfields are differentially affected. This predicts that vascular insufficiency is occuring that may be manifest as poor perfusion and hypoxia with reduced performance of parenchymal cells.

Aim A2

Significant time and dose dependent loss of neural precursor cells is observed in a lineage-specific pattern. Loss of neural precursor cells is associated with cognitive impairment and spatial memory problems.

Aim A3

Total microglial cell numbers are relatively unaffected by irradiation but sharp increases in newly-born microglia occur post-irradiation. Different phenotypic changes are seen in cells which are perivascular versus those that are not. This suggests that microglia are responding to persistent radiation-induced inflammatory changes or alterations to the microenvironment.

Aim A4

Assessment of DNA damage in the hippocampi of transgenic mice shows that mutations continue to accumulate with time after irradiation indicating the continued presence of latent DNA damage. Their structural spectrum is LET and time dependent and differs from whole brain indicating regional specialization in adaptation to genotoxicity Gene expression profiling demonstrates activation of neurotrophic and adhesion factors associated with synaptic plasticity as well as up-regulation of chemokine receptors associated with inflammatory processes.

Aim B1

Extracellular recordings from the CA1 region evaluated pyramidal cells after stimulation of CA3 axons. Input-output tests demonstrated time and dose dependent modifications to both dendritic and somatic region properties that are associated with enhanced excitability and decreased synaptic efficacy. Long-term potentiation protocols demonstrated a reduction of synaptic plasticity following irradiation that is consistent with a potential reduction of memory forming ability.

In vitro sharp electrode and patch clamp recordings showed that intrinsic membrane properties of neurons are resistant to radiation. Pharmacological manipulations then demonstrated changes in the inhibitory synapses on CA1 pyramidal cells suggesting that their hyperexcitability may be due in part to a reduction in GABAergic inhibition. Dysregulation of inhibition could, in principle, be associated with risks of seizures.

Aim B2

Diffusion Tensor Imaging visualized altered structural brain changes based on altered water diffusion properties (direction, magnitude and anisotropy). Alterations in the corpus callosum, exterior capsule and hippocampus were observed and histological comparisons are underway to determine whether early aspects of demyelination or gliosis are being detected.

Aim C1

APP23 transgenic mice recreate most of the pathological features of Alzheimer’s Disease, and most importantly, the disease progression is continuous over several months. These mice also feature prominent alterations of the vasculature, presumably one of the most vulnerable tissues in the brain after radiation. Electrophysiological measurements are showing that disease-related decreases in neuronal output and synaptic efficacy occur earlier in irradiated animals –which confirms our original hypothesis. This is consistent with the idea that that radiation may accelerate late neurogenerative changes in animals or people.

Using 3D vascular polymer cast technology combined with 3D-imaging, microvasculature changes following irradiation have been detected and are consistent with loss of vessels and an increased spacing between them. As in Aim A1 this may be indicative of reduced perfusion and function of parenchymal cells. We expect to observe neovascularization and infarcts in older APP mice and will compare the time course of this progression in control and irradiated animals to further test whether radiation causes a decreased latency in disease progression.

Aim C2

In order to assess the ability of the brain to respond to external environmental shocks and restore orderly normal function (homeostasis) we apply a controlled septic shock to cause an inflammatory reaction mediated by the periphery – nerves and immune system. The CNS response to peripheral lipopolysaccharide (LPS) exposure has long been established and typically includes changes in electrophysiology and cytokine expression. We found that in irradiated animals, the patterns of electrophysiological changes associated with reactions to LPS are complex and unlike those of either LPS or irradiation alone. We had expected the effects to be additive or synergistic. This indicates that radiation causes plastic changes to the CNS that persist for many months in the mouse and alter its program of response to septic shock. This is consistent with the idea that irradiation may potentiate the risks from late secondary insults which might include trauma or infection.

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